

SEAN McDONNELL

Software Engineer | Applied AI · Backend · Apple Platforms

Mansfield, Texas · Remote (United States) · U.S. citizen · No sponsorship required

(817) 675-9898 · nextelevenstudios@gmail.com

github.com/seanebones-lang · linkedin.com/in/mothership-ai

PROFESSIONAL SUMMARY

Software engineer who builds backend systems, applied AI, retrieval pipelines, and developer tooling in Python, Rust, and TypeScript, plus native Apple products in Swift and SwiftUI. Sole engineer behind an enterprise product line, a production SaaS suite for tattoo studios, and 12 applications on Apple's App Store. Public proof includes NextEleven Harness (Rust coding agent, 363-test suite, cross-platform CI) and PersonalRAGVault (local-first RAG with formal Hit@k / MRR / NDCG evaluation). Owns work end-to-end: architecture, implementation, tests, security review, documentation, and release. Treats CI as a release gate. Backed by 27 years of high-trust client service. Seeking remote software, backend, applied-AI, mobile, implementation, integration, or developer-tooling roles.

TECHNICAL SKILLS

Languages: Python · Rust · TypeScript/JavaScript · Swift · Kotlin · Dart · SQL · Bash

Backend & Data: FastAPI · Node.js · REST APIs · PostgreSQL/pgvector · SQLite · ChromaDB · Drizzle ORM · Redis · Celery · Supabase

AI & Retrieval: LLM APIs (Anthropic, xAI/Grok, OpenAI) · Ollama · Hugging Face fine-tuning · tool calling · MCP · multi-agent orchestration · embeddings · RAG · hybrid vector/keyword retrieval · RRF · reranking · source citations · Hit@k/MRR/NDCG · human-in-the-loop · Whisper/AssemblyAI

Apple Platforms: SwiftUI · UIKit · WidgetKit · StoreKit 2 · XCTest · Swift concurrency · URLSession · UserNotifications · CoreLocation · MapKit · AVFoundation · ActivityKit · App Intents · Keychain

Apple Delivery: App Store submission & review · version/build management · in-app purchases · widgets/extensions · privacy permissions · accessibility

Frontend & Mobile: React · Next.js · Streamlit · Jetpack Compose · Flutter · React Native · Expo · HTTP/SSE · WebSockets

Delivery & Practice: Docker · Git · GitHub Actions · automated testing · cross-platform CI · Railway · multi-tenant architecture · row-level security · threat modeling · fail-closed design

EXPERIENCE

Software Engineer — NextEleven Studios LLC / Mothership AI

September 2025 – Present · Mansfield, Texas / Remote

- Sole engineer at a two-founder company: architecture, implementation, tests, security review, release process, and documentation across an enterprise product line, a production SaaS suite, and a 12-app Apple portfolio.
- Published 12 applications on Apple's App Store (paid and free, eight categories). Owned product definition, metadata, privacy/platform requirements, submission, review feedback, versioning, and public release.
- Ship production Rust, Python, TypeScript, and Swift with automated tests and cross-platform CI as a hard release gate. NextEleven Harness carries a 363-test binary suite; PersonalRAGVault reports Hit@k, MRR, and NDCG against its own evaluation set.
- Diagnosed and fixed Windows-specific CI failures in the Rust agent (timeout and path handling) without weakening test gates; final GitHub Actions run passed all 11 jobs including Windows build, release, lint, and test.
- Fine-tuned and deployed domain-specific models on Hugging Face; built multi-provider LLM routing across hosted APIs and local Ollama inference.
- Designed agent-safety controls: no autonomous outbound action without explicit per-item approval, verifiable sources required for named entities, and evidence surfaced rather than status asserted.

Independent Software Consultant — Confidential Client Engagement

2025 · Remote

- Completed private-client contract work spanning two substantial systems: translated leadership requirements into architecture, working software, demonstrations, and technical specifications.
- Designed and built a multi-location dealership parts and inventory system (Python, FastAPI, PostgreSQL/pgvector, Redis, Next.js) with cross-location search, workflow agents, human-reviewed email classification, order/invoice flows, operator dashboards, and fail-closed external integrations.
- Implemented a second multi-component applied-AI system across Kotlin/Jetpack Compose, React Native, and Python/FastAPI, integrating retrieval, workflow orchestration, multi-device services, and explicit human-review/escalation paths.

Independent Tattoo Artist and Shop Owner

January 1998 – August 2025 · Texas · Spookshow Tattoo owner, 2006–2014

- Ran a high-trust client-service practice: consultation, requirements discovery, design iteration, expectation setting, scheduling, delivery, aftercare, difficult conversations, and long-term relationships.
- Operated Spookshow Tattoo (2006–2014): customer intake, shop standards, day-to-day operations, and time-sensitive client needs.

SELECTED PRODUCTS AND PROJECTS

Applied AI and Developer Tooling

- **NextEleven Harness v1.3.0** — Rust terminal coding agent. Multi-provider routing, sandboxed repository tools, MCP, swarm sub-agents, SQLite-backed sessions/memory, parallel workers, local HTTP/SSE, security documentation, 363-test binary suite, cross-platform CI.
- **PersonalRAGVault v1.2.0** — MIT-licensed local-first Python RAG. Ingests documents, notes, code, calendars, bookmarks; ChromaDB + FTS5 via reciprocal-rank fusion; Ollama, reranking, source citations, FastAPI/SSE + Streamlit; Hit@k, MRR, NDCG evaluation.
- **AgentGuard** — Proprietary hallucination-detection component for agent pipelines, designed for sub-50 ms inline checks.
- **AI Composer Studio v5.0** — Multi-modal generation platform (~25k LOC) with 20+ integrated models and an autonomous optimization layer.
- **Agentic AI Boilerplate** — ~6.9k LOC agent scaffold built around a human-in-the-loop review framework.

SaaS and Enterprise Systems

- **TatTools** (Linework, Render, PrimeCraft Booking) — Production SaaS suite for tattoo studios. Linework (live) does photo-to-stencil generation on tiered subscriptions; Render produces shaded/color comps; PrimeCraft handles deposit-based, zero-commission booking.
- **CRM-Desk v2.0.0** — ~17k LOC CRM on Node, React, and PostgreSQL with 14 AI-assisted features.
- **WellnessAI** — White-label FastAPI + React Native wellness platform designed against HIPAA and EU AI Act requirements.
- **Havoc HVAC SaaS** — Multi-tenant Next.js field-service platform with Twilio SMS and drag-and-drop dispatch.
- **Meetings by NextEleven** — Next.js meeting capture with Whisper/AssemblyAI transcription and LLM summarization.

Apple and Mobile

- **12 App Store applications** — Including ImanLinkPRO v2.0, If Then Reset v2.0, Pulse Social, Flight Tracker, ThriftFlip PRO, Scanner, and Tattoo Artist Helper across eight categories.
- **ImanLinkPRO v2.0** — Native SwiftUI app with StoreKit, WidgetKit, notifications, CoreLocation, networking + disk caching, encrypted local storage, app extensions, structured concurrency, and substantial XCTest coverage.
- **TrailerGuard** — Expo React Native app with BLE TPMS sensor integration, Supabase row-level security, and immutable mileage records.

Real-Time Systems

- **Sniper v1.1.0** — TypeScript, Next.js, Drizzle ORM, PostgreSQL execution platform for exchange-traded binary contracts. WebSocket market-data ingestion, fee-aware expected-value gating, Kelly position sizing, restart-safe entry tracking, and a backtesting harness.